

Matrix Computations

Lecture 02: Orthogonality and Norms

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Outline

- 1 Orthogonality, unitary and orthogonal matrices
- 2 Orthogonal expansion and orthogonal projectors
- 3 Vector and matrix norms
- 4 Matrix 2-norm and Frobenius norm

Part 1. Orthogonality, unitary and orthogonal matrices

Some notations about complex matrices

- Let $z \in \mathbb{C}$, a complex number, $z = a + ib$, $i = \sqrt{-1}$, a and b are the real and imaginary part of z , respectively.
- The complex conjugate of z : $\bar{z} = a - ib$. $\bar{\bar{z}} = z$ iff z is real
- For $A \in \mathbb{C}^{m \times n}$, its *Hermitian transpose*, $A^H \in \mathbb{C}^{n \times m}$ (in the textbook denoted as A^*). The (i, j) element of A^H is

$$A^H(i, j) = \overline{A(j, i)}$$

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix} \implies A^* = \begin{bmatrix} \bar{a}_{11} & \bar{a}_{21} & \bar{a}_{31} \\ \bar{a}_{12} & \bar{a}_{22} & \bar{a}_{32} \end{bmatrix}$$

- If $A \in \mathbb{R}^{m \times n}$, then $A^H = A^T$, the transpose of A

Inner-product and orthogonality

- Given $x \in C^m, y \in C^m$, define the **inner-product** of x and y as

$$x^H y = \sum_{i=1}^m \bar{x}_i y_i, \quad \|x\|_2 = \sqrt{x^H x}$$

- Two vectors x and y are **orthogonal** if $x^H y = 0$ (this is trivially true if either x or y is zero, so generally we exclude zero vectors when discussing orthogonality)
- A set of vectors S is orthogonal, if for any $x \in S$ and $y \in S$, and $x \neq y$, $x^H y = 0$, i.e., the vectors are pairwise orthogonal to each other
- A set of vectors S is *orthonormal*, if S is orthogonal and additionally, $\|x\|_2 = 1, \forall x \in S$ (**normalized** with Euclidean length equal to one)
- S is orthogonal implies S is linear independent (proof in class)

Unitary and orthogonal matrices

Orthogonal or unitary matrices (they are always square matrices) are essential for NLA due to their superior numerical properties (used, e.g., in QR factorization, SVD etc.)

- $Q \in C^{m \times m}$ is unitary if $Q^H Q = I_m$ — identity matrix of size m — (If Q is real, then Q is orthogonal if $Q^T Q = I_m$)
- Let $Q = [q_1, \dots, q_m] \in C^{m \times m}$. For any $v \in C^m$

$$v = \underbrace{Q Q^H}_{=I_m} v = [q_1, \dots, q_m] \begin{bmatrix} q_1^H v \\ \vdots \\ q_m^H v \end{bmatrix} = \sum_{i=1}^m (q_i^H v) q_i$$

This is an (orthogonal) **expansion** of v in terms of a given orthonormal basis $[q_1, \dots, q_m]$ of C^m

Part 2. Orthogonal expansion and orthogonal projectors

Orthogonal expansion and orthogonal projector

- For $n < m$, we write

$$v = \sum_{i=1}^n (q_i^H v) q_i + r, \quad r = \sum_{i=n+1}^m (q_i^H v) q_i$$

- Let $\mathcal{S} = \text{span}\{q_1, \dots, q_n\}$, an n -dimensional linear subspace of C^m
- Then

$$\sum_{i=1}^n (q_i^H v) q_i \in \mathcal{S}, \quad r \in \mathcal{S}^\perp = \text{span}\{q_{n+1}, \dots, q_m\}$$

$\mathcal{S}^\perp$ is the orthogonal complement of $\mathcal{S}$

$$\mathcal{S} \oplus \mathcal{S}^\perp = C^m$$

Orthogonal expansion and orthogonal projector

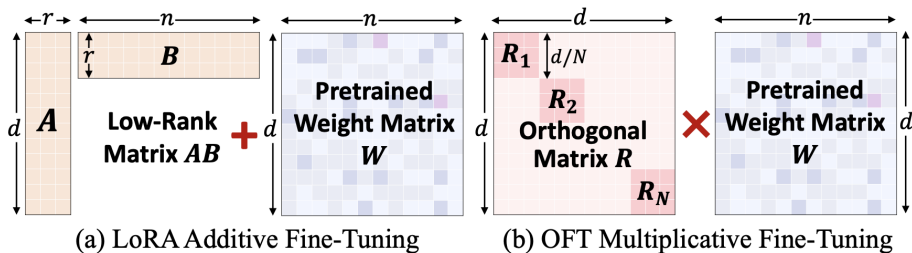
- Let $Q_1 = [q_1, \dots, q_n]$, and $Q = [Q_1, Q_2]$, then

$$\sum_{i=1}^n (q_i^H v) q_i = Q_1 Q_1^H v, \quad r = Q_2 Q_2^H v = (I - Q_1 Q_1^H) v$$

- Additionally, $I = Q_1 Q_1^H + Q_2 Q_2^H$
- $Q_1 Q_1^H$ and $Q_2 Q_2^H$ are **orthogonal projectors** onto the linear subspaces $\mathcal{S}$ and $\mathcal{S}^\perp$, respectively
- It's easy to verify that $Q_1 Q_1^H$ and $Q_2 Q_2^H$ are Hermitian matrices, and

$$(Q_1 Q_1^H)^2 = Q_1 Q_1^H, \quad (Q_2 Q_2^H)^2 = Q_2 Q_2^H$$

Low-rank and orthogonal matrices in fine-tuning of LLM



Controlling text-to-image diffusion by orthogonal finetuning. Qiu et. al., NeurIPS, 2023

Parameter Efficient Quasi-Orthogonal Fine-Tuning via Givens Rotation. Ma et. al., ICML, 2024

Many variations of the above ideas!

Part 3. Vector and matrix norms

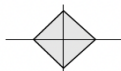
Vector and matrix norms

- The concept of the size or magnitude of a vector or distance between two vectors is captured by vector norms
 - This notion is essential for measuring the size of errors and residuals (in LS problems, for example)
- A **norm** is a real function $\| \cdot \|: C^m \rightarrow R_+$ (the set of nonnegative real numbers), satisfying
 - 1 $\|x\| \geq 0$, and $\|x\| = 0$ only if $x = 0$
 - 2 $\|x + y\| \leq \|x\| + \|y\|$ (triangle inequality)
 - 3 $\|\alpha x\| = |\alpha| \|x\|, \forall \alpha \in C$
- Similarly, norms can be defined for matrices as well

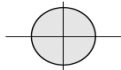
Vector norms

- Some examples of vector norms, and the corresponding closed unit ball $\{x \in C^m, \|x\| \leq 1\}$

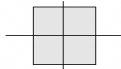
$$\|x\|_1 = \sum_{i=1}^m |x_i|,$$



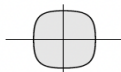
$$\|x\|_2 = \left(\sum_{i=1}^m |x_i|^2 \right)^{1/2} = \sqrt{x^* x},$$



$$\|x\|_\infty = \max_{1 \leq i \leq m} |x_i|,$$



$$\|x\|_p = \left(\sum_{i=1}^m |x_i|^p \right)^{1/p} \quad (1 \leq p < \infty).$$



- What if $p < 1$ in $\|\cdot\|_p$?

Matrix norms induced by vector norms

- Let $A \in C^{m \times n}$, and we define an induced matrix norm (or **operator norm**) by

$$\|A\| = \max_{x \neq 0} \frac{\|Ax\|}{\|x\|}$$

Note. In general, the norm for $\|Ax\|$ and $\|x\|$ can be different. For example, one is 1-norm and the other is 2-norm

- It is easy to check $\|\cdot\|$ thus defined satisfies all three norm properties, and additionally

$$\|Ax\| \leq \|A\|\|x\|, \quad \|AB\| \leq \|A\|\|B\|$$

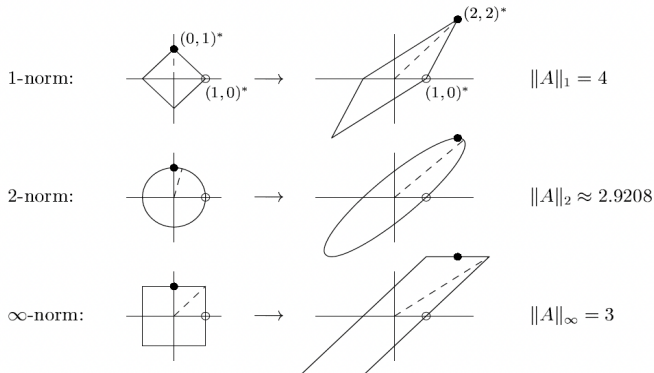
the so-called *compatibility conditions* (between the vector norms and the induced matrix norm) (Try this proof yourself)

Visualize matrix norms

- Consider

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}$$

and its action on the unit balls of three norms



Example: induced matrix p -norms of a diagonal matrix

- Let $D = \text{diag}(d_1, \dots, d_m)$, then

$$\|Dx\|_p = \left(\sum_{i=1}^m |d_i x_i|^p \right)^{1/p} \leq \max_{1 \leq i \leq m} |d_i| \left(\sum_{i=1}^m |x_i|^p \right)^{1/p}$$

which implies $\|D\|_p \leq \max_{1 \leq i \leq m} |d_i|$

- Now pick $x \in C^m$ with $x_k = 1$ for the index k satisfies $|d_k| = \max_{1 \leq i \leq m} |d_i|$, and all other entries equal to zero. Then

$$\|Dx\|_p = |d_k| = \max_{1 \leq i \leq m} |d_i| \|x\|_p$$

- Conclusion

$$\|D\|_p = \max_{1 \leq i \leq m} |d_i|$$

Explicit formula for matrix 1-norms and ∞ -norm

- 1-norms: let $A = [a_1, \dots, a_n]$, then

$$\|A\|_1 = \max_{1 \leq i \leq n} \|a_i\|_1$$

- ∞ -norm, let

$$\begin{bmatrix} A_1^T \\ \vdots \\ A_m^T \end{bmatrix}$$

then

$$\|A\|_\infty = \max_{1 \leq i \leq m} \|A_i\|_1$$

(Try this proof yourself)

Part 4. Matrix 2-norm and Frobenius norm

Matrix 2-norm and Frobenius norm

- Ferdinand Georg Frobenius (1849 – 1917)
- Let $A \in \mathbb{C}^{m \times n}$, and

$$\|A\|_2 = \max_{x \neq 0} \frac{\|Ax\|_2}{\|x\|_2}$$

which turns out to be the largest **singular value** of A , as we will show later

- Let $A = [a_{ij}] \in \mathbb{C}^{m \times n}$, its Frobenius norm, which is easy to compute, is defined as ($\text{tr}(\cdot)$ is the trace of a matrix)

$$\|A\|_F = \left(\sum_{i,j=1}^m |a_{ij}|^2 \right)^{1/2} = \sqrt{\text{tr}(A^H A)} = \sqrt{\text{tr}(A A^H)}$$

- $\|\cdot\|_F$ is not an induced matrix norm since $\|I\|_F \neq 1$ if $m = n > 1$

Unitary invariance of 2-norm and Frobenius norm

- Let Q be unitary,

$$\|Qx\|_2^2 = (Qx)^H Qx = x^H Q^H Qx = \|x\|_2^2$$

- It can be proved also, for any two unitary matrices P and Q

$$\|PAQ\|_2 = \|A\|_2, \quad \|PAQ\|_F = \|A\|_F$$

This is to say 2-norm and Frobenius norm are *unitarily invariant*

- Because of the unitary invariance, both 2-norm and Frobenius norm are certain functions of the singular values of the matrix (Try this proof yourself using SVD)

Frobenius norm is compatible

- The proof is simple, let $C = AB$, a_i^H , i -th row of A , and b_j , j -th column of B , by way of Cauchy-Schwarz

$$\begin{aligned}\|AB\|_F^2 &= \sum_{i=1}^n \sum_{j=1}^m |c_{ij}|^2 \\ &\leq \sum_{i=1}^n \sum_{j=1}^m (\|a_i\|_2 \|b_j\|_2)^2 \\ &= \sum_{i=1}^n (\|a_i\|_2)^2 \sum_{j=1}^m (\|b_j\|_2)^2 = \|A\|_F^2 \|B\|_F^2.\end{aligned}$$

- Actually, a sharper bound $\|AB\|_F \leq \|A\|_2 \|B\|_F$ can be proved which implies compatibility, since $\|A\|_2 \leq \|A\|_F$ (which can be proved by Cauchy-Schwarz, try this proof yourself)

Frobenius norm and block matrices

- If $A = [a_1, \dots, a_n]$, then

$$\|A\|_F^2 = \|a_1\|_2^2 + \dots + \|a_n\|_2^2$$

- A similar formula exists if we partition A row-wise
- Or more generally, if we write $A = [A_{ij}]$ as a block matrix with each A_{ij} a submatrix of A

$$\|A\|_F^2 = \sum_{i,j} \|A_{ij}\|_F^2$$